ECO 202 — Practice Exam 4 — Version B

Statistics and Data Analysis for Economics | Princeton University

Fall 2026 | Classes 14–18 | 100 points

Name: ______________________________ Student ID: ________________

Timing. Target workload: 70 minutes (10 minutes multiple choice; 60 minutes written work). Follow announced start/end times within the 80-minute meeting. For practice, set a 70-minute timer.

Conditions. No books, notes, calculators, AI, or other electronics; University-authorized accommodations apply. Fractions, radicals, and equivalent exact expressions are acceptable. Show reasoning in written answers. Data are teaching examples unless identified otherwise.

Points. Questions 1–4: one answer A–D, 5 points each, no negative marking. Questions 5–8: 20 points each; subpart points are shown. Label any continuation clearly.

Multiple choice (20 points; about 10 minutes)

1. A test rejects a null hypothesis that is actually true. This is:

A. A Type II error.

B. Power.

C. Proof that the procedure is invalid.

D. A Type I error.

2. Before/after measurements on the same independently sampled households are positively correlated. Ignoring that pairing in the standard error of the mean difference ordinarily:

A. Makes the observed mean difference causal.

B. Understates its variance because both columns have the same size.

C. Leaves the standard error unchanged.

D. Overstates its variance by omitting a positive covariance subtraction.

3. There are 20 successes in 100 independent trials. For a null-based Normal test of $p = 1/4$, which standard error is appropriate?

A. $\sqrt{(1/5)(4/5)/100}$.

B. $\sqrt{(1/4)(3/4)/100}$.

C. $\sqrt{(1/4)(3/4)}$.

D. $(1/4)(3/4)/100$.

4. Adding pre-treatment experience to a wage regression changes the education slope from 0.5 to 0.4. Which conclusion is justified by that change alone?

A. The causal effect is 0.4.

B. The omitted-variable bias was exactly 0.1.

C. The adjusted coefficient is necessarily closer to the causal effect.

D. The two fitted models make different conditional comparisons.

5. Two populations and a difference in means (20 points; about 15 minutes)

Independent random samples of 100 households in Region A and 100 in Region B are used to compare mean numbers of cars. Both population standard deviations are known to equal 1 car. The observed means are $\bar{x}_A = 3/2$ and $\bar{x}_B = 6/5$. Let $\Delta = \mu_A - \mu_B$ and use a large-sample Normal approximation.

The standard error for the difference is $\sqrt{\sigma_A^2/n_A + \sigma_B^2/n_B}$. For $Z \sim \mathsf{N}(0,1)$, $\mathbb{P}(Z > 3/\sqrt{2}) = 0.01695$. Use the two-sided 95% critical value 1.96 and the upper-sided 5% critical value 1.645.

(a) (4 points) Calculate the estimated difference and its standard error.

(b) (6 points) Test $H_0 : \Delta = 0$ against $H_a : \Delta \neq 0$ at 5%, including the statistic, p-value and contextual conclusion.

(c) (5 points) Construct the approximate 95% interval and explain its agreement with the test.

(d) (5 points) A separate upper-sided procedure rejects when $\widehat{\Delta} > 1.645\,\mathrm{SE}$. Under a planning model with fixed SE and $\widehat{\Delta} \sim \mathsf{N}(\Delta, \mathrm{SE}^2)$, what are power and Type II error probability when the true $\Delta = 1.645\,\mathrm{SE}$? Explain without a probability table.

6. An observational threshold comparison (20 points; about 15 minutes)

Two independent random samples of 100 households each are drawn from distinct large populations: those living near a train station and those living farther away. In the near group, 80 households spend below a specified monthly transport-cost threshold; in the far group, 50 do. Location was chosen by households, not randomly assigned. Let $\Delta_p = p_{\text{near}} - p_{\text{far}}$.

Use the unpooled estimate-based standard error

$$\widehat{\text{SE}} = \sqrt{\frac{\widehat{p}_{\text{near}}(1-\widehat{p}_{\text{near}})}{100} + \frac{\widehat{p}_{\text{far}}(1-\widehat{p}_{\text{far}})}{100}},$$

and the Normal critical value 1.96. A proposed two-sided Wald test uses this *same* standard error; no pooled test is requested.

(a) (4 points) Calculate the two proportions and their difference. Give the difference in percentage points.

(b) (6 points) Calculate the standard error and construct an approximate 95% confidence interval.

(c) (5 points) Determine the matching 5% test decision for $\Delta_p = 0$, justifying the comparison without a calculator. Check observed successes and failures in both groups.

(d) (5 points) Does statistical significance establish that moving closer to a station causes a household to cross the threshold, or that the difference is economically important? Explain what else is needed for each conclusion.

7. Conditional information and overall wealth (20 points; about 15 minutes)

In a fictional population, X is the number of phones a household owns and Y is net wealth in tens of thousands of dollars. You know:

x	0	1	2
$\mathbb{P}(X = x)$	1/4	1/2	1/4
$\mathbb{E}[Y \mid X = x]$	4	10	12

Within the $X = 1$ group, $Y = 8$ and $Y = 12$ each have conditional probability 1/2.

(a) **(4 points)** Calculate $\mathbb{P}(X = 1, Y = 12)$ and verify the reported conditional mean for $X = 1$.

(b) **(6 points)** Find overall mean wealth using iterated expectations. Explain why an unweighted average of the three displayed means is generally wrong.

(c) **(5 points)** Is the regression function $f(x) = \mathbb{E}[Y \mid X = x]$ linear? Would a best linear predictor necessarily equal all three displayed conditional means? Explain.

(d) **(5 points)** Define $\varepsilon = Y - f(X)$. Calculate its two possible values within $X = 1$ and its conditional mean. Does the wealth pattern establish that buying a phone increases wealth? Explain briefly.

8. A population line and an estimated line (20 points; about 15 minutes)

For a teaching population, X is commute distance in kilometers and Y is weekly transit trips. The population moments are $\mathbb{E}[X] = 2$, $\mathbb{E}[Y] = 5$, $\mathrm{Var}(X) = 4$, and $\mathrm{Cov}(X, Y) = -2$. For the best linear predictor, use

$$\beta_1 = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)}, \qquad \beta_0 = \mathbb{E}[Y] - \beta_1 \mathbb{E}[X].$$

An independent random sample from this population produces the fitted line $\widehat{y} = 5.8 - 0.4x$ and a valid robust standard error of 0.1 for its slope. A large-sample Normal approximation applies. Use 1.96 and $\mathbb{P}(Z > 1) = 0.1587$ for $Z \sim \mathsf{N}(0, 1)$.

(a) (5 points) Calculate the population best-linear-prediction coefficients. Explain why the sample coefficients need not equal them.

(b) (4 points) Calculate the sample line's prediction at $x = 3$ and interpret its slope with units.

(c) (7 points) Construct a 95% confidence interval from the sample slope and standard error. Test the benchmark $H_0 : \beta_1 = -1/2$ against $H_a : \beta_1 \neq -1/2$ at 5%, including the statistic, p-value and decision.

(d) (4 points) A planner says: "The negative slope proves that relocating any household one kilometer farther away reduces its weekly transit trips by the slope amount." Identify the unsupported causal and individual-level conclusions.